

flspcell_slow_1p35V_125C Library

Cell Groups
DFRBPQ

DFRBPQ



*flspcell_slow_1p35V_125C Cell Library: Process
flspcell_slow_1p35V_125C, Voltage 1.35, Temp 125.00*

Truth Table

INPUT			OUTPUT
CLK	D	RESET_B	Q
R	0	1	0
R	1	1	1
x	x	0	0
x	x	1	IQ

Footprint

Cell Name	Area
sg13g2_dfrbpq1arx_1	48.98880

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	CLK	D	RESET_B	Q
sg13g2_dfrbpq1arx_1	0.02170	0.01499	0.02651	0.30000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_dfrbpq1arx_1	17972.40000	20851.10000	24585.40000

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RR)	0.01860	0.00100	0.37225	0.32940	0.06480	0.64599	2.50740	0.30000	1.51899

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RF)	0.01860	0.00100	0.41158	0.32940	0.06480	0.68819	2.50740	0.30000	1.44231
	RESET_B->Q (FF)	0.01860	0.00100	0.48642	0.32940	0.06480	0.80290	2.50740	0.30000	1.74651

Constraint Information

Constraints(ns) for CLK rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.11826	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for CLK falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.12146	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.05379	1.26300	1.26300	-0.18079	2.50740	2.50740	-0.23612
	setup	CLK (R)	0.01860	0.01860	0.47681	1.26300	1.26300	0.55047	2.50740	2.50740	0.56965

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.02690	1.26300	1.26300	-0.17269	2.50740	2.50740	-0.27449
	setup	CLK (R)	0.01860	0.01860	0.51104	1.26300	1.26300	0.62602	2.50740	2.50740	0.70837

Constraints(ns) for RESET_B rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	recovery	CLK (R)	0.01860	0.01860	0.11737	1.26300	1.26300	0.25904	2.50740	2.50740	0.36894
	removal	CLK (R)	0.01860	0.01860	-0.10759	1.26300	1.26300	-0.25095	2.50740	2.50740	-0.36009

Constraints(ns) for RESET_B falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	RESET_B ()	0.01860	0.00000	0.11185	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.17451	0.32940	0.06480	0.17667	2.50740	0.30000	0.23789

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.19679	0.32940	0.06480	0.20240	2.50740	0.30000	0.26483
	RESET_B	0.01860	0.00100	0.18329	0.32940	0.06480	0.18471	2.50740	0.30000	0.21001

Passive power(pJ) for CLK rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.04327	0.32940	0.04626	2.50740	0.10505

Passive power(pJ) for CLK falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.07330	0.32940	0.07657	2.50740	0.13932

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.03077	0.32940	0.03394	2.50740	0.09261
	(D * !RESET_B * * !Q)	0.01860	0.04327	0.32940	0.04626	2.50740	0.10505
	(!D * RESET_B * !Q)	0.01860	0.04538	0.32940	0.04845	2.50740	0.10728
	(!D * !RESET_B * * !Q)	0.01860	0.05118	0.32940	0.05411	2.50740	0.11294

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.07330	0.32940	0.07657	2.50740	0.13932
	(D * RESET_B * !Q)	0.01860	0.07908	0.32940	0.08231	2.50740	0.14510
	(D * !RESET_B * * !Q)	0.01860	0.03834	0.32940	0.04168	2.50740	0.10248
	(!D * RESET_B * Q)	0.01860	0.21083	0.32940	0.18845	2.50740	0.24856
	(!D * RESET_B * !Q)	0.01860	0.03347	0.32940	0.03675	2.50740	0.09755
	(!D * !RESET_B * * !Q)	0.01860	0.02969	0.32940	0.03297	2.50740	0.09374

Passive power(pJ) for D rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.06901	0.32940	0.06900	2.50740	0.08389

Passive power(pJ) for D falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.06887	0.32940	0.06943	2.50740	0.09246

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.06640	0.32940	0.06676	2.50740	0.08872
	(!CLK * RESET_B)	0.01860	0.11284	0.32940	0.11324	2.50740	0.13586
	(!CLK * !RESET_B)	0.01860	0.06901	0.32940	0.06900	2.50740	0.08389

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.06887	0.32940	0.06943	2.50740	0.09246
	(!CLK * RESET_B)	0.01860	0.09113	0.32940	0.09158	2.50740	0.11575
	(!CLK * !RESET_B)	0.01860	0.05193	0.32940	0.05207	2.50740	0.06783

Passive power(pJ) for RESET_B rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.01320	0.32940	0.01347	2.50740	0.03235

Passive power(pJ) for RESET_B falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.03348	0.32940	0.03233	2.50740	0.06316

Passive power(pJ) for RESET_B rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.01320	0.32940	0.01347	2.50740	0.03235
	(CLK * !D * !Q)	0.01860	0.00653	0.32940	0.00657	2.50740	0.00662
	(!CLK * D * !Q)	0.01860	0.05719	0.32940	0.05707	2.50740	0.08480
	(!CLK * !D * !Q)	0.01860	0.01037	0.32940	0.01041	2.50740	0.01046

Passive power(pJ) for RESET_B falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.21799	0.32940	0.21811	2.50740	0.27504
	(CLK * !D * !Q)	0.01860	-0.00491	0.32940	-0.00569	2.50740	-0.00595
	(!CLK * D * !Q)	0.01860	0.03348	0.32940	0.03233	2.50740	0.06316
	(!CLK * !D * !Q)	0.01860	-0.00934	0.32940	-0.01001	2.50740	-0.01023